2. This problem deals with compactness

a. Prove that a finite union of compact subsets of a metric space is compact

Definitions: compact: every open cover has a finite subcover

open cover: $\mathcal{U}$ is an open cover of E if each point in E belongs to at least one set in $\mathcal{U}$

subcover: any subfamily of $\mathcal{U}$ that also covers E

Proof: let $A_1, \ldots, A_n$ be compact subsets of a metric space (X, d) for a finite $n \in \mathbb{N}$.

let $A = A_1 \cup \ldots \cup A_n$.

let $O = \{O_i\}$ for some index i be any open cover of A

Since O covers all of A it must cover each $A_1, \ldots, A_n$ as well

Since $A_1, \ldots, A_n$ are compact and covered, they each must have finite subcovers $\{\{a_i\}, \ldots, \{a_n\}\}$

let $V = a_1 \cup \ldots \cup a_n$ be a finite subcover of $\mathcal{U}$ that covers all of A.

Since $\mathcal{U}$ was arbitrary, A is compact.

b. Prove that the subset $\{0, 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\}$ is compact directly from the definition

Let $X = \{0, 1, \frac{1}{2}, \ldots\}$.

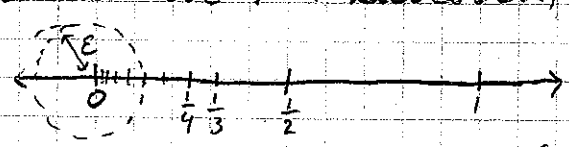
Since there exists some arbitrary open cover over A, there must be some subcover containing 0. Since E is open it cannot equal $\{0\}$, therefore it must cover some interval $(-\varepsilon, \varepsilon)$.

If $\frac{1}{n} < \varepsilon$, then $\frac{1}{n} \in E$ for any $n \in \mathbb{N}$.

let M be the largest integer such that $\frac{1}{m} \notin (-\varepsilon, \varepsilon)$.

Then a subcover can be constructed with at most M sets covering $[\frac{1}{m}, 1] \cap X = \{\frac{1}{m}, \frac{1}{m-1}, \ldots, 1\}$

Therefore we have $M+1$ subcovers, a finite amount



can see that infinite part of sequence will always be contained is subcover covering 0

c. Construct a bounded set of real numbers with exactly three limit points

Definition: a point x is a limit point of E if every neighborhood of x intersects E in a point that is not x. For example, $\{3\}$ does not have a limit point but every point in $[0,1]$ is a limit point.

Recall from 2b that $\{0, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\}$ is a set with exactly one limit point.

So to construct a set with 3 limit points, use the same principle

$$A = \{a + \tfrac{1}{n} \mid a \in \{0,1,2\} \text{ and } n \in \mathbb{N}\}$$

This set is bounded by 0, 3:

the smallest element will be $a = 0 + \lim_{n\to\infty} \frac{1}{n} = 0 = 0$

the largest element will be $a = 2$, $n = 1$ for $2 + 1 = 3$

This set has 0, 1, 2 as limit points:

WLOG, let us prove that 0 is a limit point.

take the largest $n \in \mathbb{N}$ s.t. $\forall \varepsilon$, $\exists$ a $\text{ball}_\varepsilon(0)$

WTS $B_\varepsilon(0) \cap A \neq 0$.

let $N = \frac{1}{\varepsilon}$, then $\forall n > N$, $\frac{1}{n} \in \text{ball}_\varepsilon(0)$ and $\frac{1}{n} \in A$ by the way A is defined.

Since ε was arbitrary, 0 is a limit point.

d. Prove directly from the definition (ie, without using Heine-Borel or Cantor intersection lemma) that $[0,1] \subset \mathbb{R}$ is compact.

Let $E = [0,1]$.

$E \subset \mathbb{R}$ and $\mathbb{R}$ is compact, so we want to show that every cell F in $\mathbb{R}$ is compact.

Pf: Assume F is not compact. Then there exists an open cover $\mathcal{U}$ of F, no finite subfamily of which covers F. Let δ denote the diameter of F.

F is the union of 2 cells with diameter $\delta/2$. One of these cells, F_1, cannot be covered by finitely many sets from $\mathcal{U}$.

Likewise, F_1 contains a cell F_2 of diameter $\delta/4$ which cannot be covered by finitely many sets from $\mathcal{U}$.

Continuing in this fashion, we obtain a sequence (F_n) of cells such that

$$F_1 \supseteq F_2 \supseteq F_3 \supseteq \ldots$$

where each F_n has diameter $\delta/2^n$ and F_n cannot be covered by finitely many sets from $\mathcal{U}$.

By Thm 13.10 from Ross, the intersection $\bigcap_{n=1}^{\infty} F_n$ contains a point x_0 belonging to some set U_0 in $\mathcal{U}$.

Since U_0 is open, $\exists\, r > 0$ so that $\{x \in \mathbb{R} \mid d(x, x_0) < r\} \subseteq U_0$.

It follows that $F_n \subseteq U_0$ provided $\delta/2^n < r$, but this contradicts our assertion that F_n cannot be covered by finitely many sets from $\mathcal{U}$.

– Taken from Ross, pages 90-91